

Chem E 467
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Homework 6
Due: 5/13/26

1) Given $\mu = 0.04 \text{ hr}^{-1}$ $\mu_{\max} = 0.8 \text{ hr}^{-1}$
 $Y_{X/S}^m = 0.4 \frac{\text{g cell}}{\text{g can oil}}$ $m = 0.05 \frac{\text{g can oil}}{\text{g cell} \cdot \text{hr}}$
 $K_s = 5 \frac{\text{g}}{\text{L}}$ $q_p = 0.1 \frac{\text{g prop acid}}{\text{g cell} \cdot \text{hr}}$
 $Y_{P/S} = 0.8 \frac{\text{g prop acid}}{\text{g can oil}}$ $S_0 = 200 \frac{\text{g}}{\text{L}}$

chemostat w/o recycle - $D = \mu$ so $D = 0.04 \text{ hr}^{-1}$
 canola oil ($\text{C}_{18}\text{H}_{32}\text{O}_2$) $\text{MW} = 280.4 \text{ g/mol}$
 prop acid ($\text{C}_3\text{H}_6\text{O}_2$) $\text{MW} = 74.1 \text{ g/mol}$
 feed solution has can oil concentr. of 200 g/L

$$\mu = \frac{\mu_{\max} S}{K_s + S} \rightarrow S = \frac{K_s \mu}{\mu_{\max} - \mu}$$

$$S = \frac{(5)(0.04)}{0.8 - 0.04} = \frac{0.2}{0.76} = 0.26316$$

$$D(S_0 - S) = C_x \left(\frac{\mu}{Y_{X/S}^m} + m + \frac{q_p}{Y_{P/S}} \right)$$

$$\text{Solve } C_x = \frac{D(S_0 - S)}{\frac{\mu}{Y_{X/S}^m} + m + \frac{q_p}{Y_{P/S}}}$$

$$C_x = \frac{0.04 (200 - 0.26316)}{\frac{0.04}{0.4} + 0.05 + \frac{0.1}{0.8}} = \frac{7.9895}{0.275} = 29.053 \text{ g/L}$$

a,

~~0.1 (29.053)~~

$$r_p = (0.1)(29.053) = 2.905 \text{ g/(L-hr)}$$

b,

$$Dc_p = q_p C_x \rightarrow$$

$$C_p = \frac{q_p C_x}{D}$$

~~0.04 (29.053)~~
~~0.04~~

$$C_p = \frac{2.905}{0.04} = 72.63 \text{ g/L}$$

2. Given $S_0 = 100 \text{ g/L}$ $\mu_{max} = 0.8 \text{ hr}^{-1}$

SS $\rightarrow$ Dzu

glycerol ($\text{C}_3\text{H}_8\text{O}_3$) MW = 92 g/mol
Dant ($\text{C}_4\text{H}_{10}\text{O}$) MW = 74 g/mol

a) $\mu = \frac{\mu_{max} S}{K_s + S} \rightarrow D = \mu$

$$K_s = \frac{S(\mu_{max} - D)}{D}$$

$$\frac{5.0 (0.8 - 0.5)}{0.5} = 3.0 \text{ g/L}$$

$$\frac{2.1 (0.8 - 0.7)}{0.7} = 3.0 \text{ g/L}$$

$$\boxed{3.0 \text{ g glycerol/L}}$$

b) $\frac{1}{Y_{obs}} = \frac{1}{Y_m} + \frac{m}{D} \rightarrow \text{linear form } y = b + mx$

$Y = \frac{1}{Y_{obs}}$, $X = \frac{1}{D}$, slope = m, intercept = $\frac{1}{Y_m}$

$Y_{obs} = C_4 / (C_0 - C_5) = 0.285$

$D = 0.1 \rightarrow X = 10, Y = 3.51$

$D = 0.7 \rightarrow X = 1.429, Y = 2.64$

$$\frac{(3.51 - 2.64)}{(10 - 1.429)} = 0.101$$

intercept $2.64 - 0.101 (1.429) = 2.50$

$$\boxed{\frac{1}{Y_m} = 2.5}$$

$$\frac{1}{2.5} = 0.400$$

$$\boxed{Y_m = 0.400 \frac{\text{g cell}}{\text{g glycerol}}}$$

3) Given: $V_L = 50,000 \text{ L}$

$V_T = 65,000 \text{ L}$

$$q_p = 0.2 \text{ g naph} / \text{g cell hr}$$

$$\mu = 0.15 \text{ hr}^{-1}$$

$$Y_{X/S} = 0.4 \text{ g cell} / \text{g gluc}$$

$$Y_{P/S} = 0.25 \text{ g ~~gluc~~ naph} / \text{g gluc}$$

34' exerts
1 atm

$$O_2 \text{ xfr } K_{La} = 30 \text{ mmol} / (\text{L} \cdot \text{atm} \cdot \text{min})$$

Biomass $(CH_{1.8}O_{0.5}N_{0.25})$

probe @ bottom
of ferm, the
dissolved $O_2 = 2\%$
air saturation
11 ← calib

Glucose $C_6H_{12}O_6 \rightarrow MW = 180$

Naph $C_{10}H_8 \rightarrow MW = 128.2$

Nitrogen source NH_3

Ferm HP = 2 atm

2nd dissolv. O_2

probe @ bottom
reads 30%

head is 20' w/ aerated conditions w/ 15% gas hold up V
dissolved

Inlet $g O_2$ @ bottom = 0.21

exit gas $O_2 = 0.16$

$$Y_{O_2} = 0.21$$

$$Y_{O_2} = 0.16$$

Dissolved O_2 readings converted using 2.0 atm

15% gas hold-up reduces effective head

$$O_2 \text{ xfer capacity : } OUR = k_{La} (\Delta P_{O_2})_{LM}$$

log mean O_2
driving force

$$P_{top} = 2.0 \text{ atm}$$

$$h_{eff} = 20' (1 - 0.15) = 17' H_2O$$

$$\Delta P = \frac{17}{31.1 \text{ atm}} = 0.50 \text{ so } P_{bottom} = 2.0 + 0.5 = 2.5$$

top pressure = 2.0 atm bottom pressure = 2.5 atm

$$P_{O_2,b}^* = Y_{O_2,b} P_{bottom} \rightarrow (0.21)(2.5) = 0.525 \text{ atm}$$

$$P_{O_2,t}^* = Y_{O_2,t} P_{top} \rightarrow 0.16 \times 2.0 = 0.320 \text{ atm}$$

Both probes @ 100% air sat @ 2.0 atm

$$P_{O_2, \text{sat}} = (0.21)(2.00) = 0.420$$

bottom probe reads 30% air sat.

$$P_{O_2, b} = (0.30)(0.420) = 0.126 \text{ atm}$$

top @ 27% air sat

$$P_{O_2, t} = (0.27)(0.420) = 0.1134 \text{ atm}$$

Bottom driving force

$$\Delta P_b = P_{O_2, b} - P_{O_2, L, b} \rightarrow 0.525 - 0.126 = 0.399 \text{ atm}$$

$$\text{Top } \Delta P_t = P_{O_2, t} - P_{O_2, L, t} \rightarrow 0.320 - 0.1134 = 0.2066 \text{ atm}$$

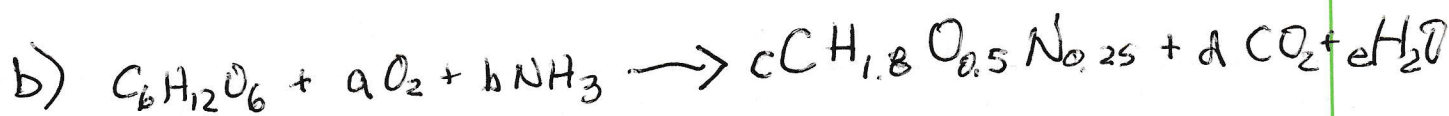
log mean driving force

$$(\Delta P_{O_2})_{LM} = \frac{\Delta P_b - \Delta P_t}{\ln(\Delta P_b / \Delta P_t)} \rightarrow \frac{0.399 - 0.2066}{\ln(0.399 / 0.2066)} = \frac{0.1924}{\ln(1.93)}$$

$$\frac{0.1924}{0.6580} = 0.2924 \text{ atm}$$

$$OVR = (30)(0.2924) = 8.772$$

$$\boxed{8.772 \frac{\text{mmol } O_2}{L \cdot \text{min}}}$$



$$\begin{aligned} \text{MW 1 mole biomass} &\rightarrow 12 + 1.8 + 0.5(16) + 0.25(14) \\ &\rightarrow 25.3 \text{ g} \end{aligned}$$

$$1 \text{ mole glucose } 180 \text{ g gluc} \rightarrow 180 \times (0.4) = 72 \text{ g cell}$$

$$\text{Conv. cell mass to moles } C = \frac{72}{25.3} = 2.846$$

growth rxn balance

$$\underline{N} \rightarrow b = 0.25(2.846) = 0.7115$$

$$\underline{C} \rightarrow 6 = c + d \quad d = 6 - 2.846 = 3.154 \quad 2.1345$$

$$\underline{H} \rightarrow 12 + 3b = 1.8c + 2e \rightarrow 12 + 3(0.7115) = 14.1345 = 1.8(2.846) + 2e \rightarrow 9.0117 = 2e$$

$$\cancel{e = 4.506} \quad e = 4.506$$

$$\underline{O} \rightarrow 6 + 2a = 0.5c + 2d + e$$

$$6 + 2a = 0.5(2.846) + 2(3.154) + 4.506$$

$$a = 3.119$$

so O_2 req / g of glnc. for growth

$$\frac{3.119(1000)}{180} = 17.33 \frac{\text{mmol } O_2}{\text{g glnc.}}$$

$$SVR_{\text{growth}} = \frac{\mu C_x}{Y_{X/S}} = \frac{0.15 C_x}{0.4} \rightarrow 0.375 C_x$$

$$OVR_{\text{growth}} = 17.33(0.375 C_x) \rightarrow 6.499 C_x \frac{\text{mmol } O_2}{\text{L-hr}}$$

Naph production



$$\text{naph mass} = 180(0.25) = 45 \text{ g naph}$$

$$f = \frac{45}{128.2} = 0.351$$

Product rxn bal.

$$C \rightarrow 6 = 10f + d \rightarrow d = 6 - 10(0.351) = 2.49$$

$$H \rightarrow 12 = 8f + 2e \rightarrow e = \frac{12 - 8(0.351)}{2} = 4.60$$

$$O \rightarrow 6 + 2a = 2d + e \rightarrow 6 + 2a = 2(2.49) + 4.60 = 1.788$$

S_0 naph form req. $1.788 \frac{\text{mol } O_2}{\text{mol gluc}}$

$$O_2 / g \text{ gluc} \quad \frac{1.788(1000)}{180} = 9.933$$

$$SUR_{\text{product}} \quad \frac{0.2 C_x}{0.25} = 0.80 C_x$$

$$OUR_{\text{prod}} \quad (9.933)(0.8 C_x) = 7.946 C_x \frac{\text{mmol } O_2}{\text{L} \cdot \text{hr}}$$

$$OUR_{\text{total}} \quad 6.499 C_x + 7.946 C_x = 14.445 C_x$$

$$14.445 \frac{\text{mmol } O_2}{\text{L} \cdot \text{hr}} \quad \frac{1 \text{ hr}}{60 \text{ min}} = 0.2408 \frac{\text{mmol } O_2}{\text{L} \cdot \text{min}}$$

$$8.772 = 0.2408 C_x \rightarrow C_x = \boxed{36.43 \frac{\text{g cells}}{\text{L}}}$$

c) Gluc conc. in feed stream

$D = \mu = 0.15$ because residual gluc is negligible

$$D(S_0 - S) \approx DS_0$$

$$DS_0 = SUR_{\text{growth}} + SUR_{\text{product}}$$

$$DS_0 = \frac{\mu C_x}{Y_{x/s}} + \frac{q_p C_x}{Y_{p/s}} \rightarrow 0.375 C_x + 0.80 C_x = 1.175 C_x$$

$$DS_0 = 1.175 C_x \rightarrow S_0 = \frac{1.175 C_x}{D} \rightarrow \frac{1.175(36.43)}{0.15}$$

$$\boxed{S_0 = 285.4 \frac{\text{g glucose}}{\text{L}}}$$

d) Volumetric productivity r_p tot ferm V

$$r_p = q_p C_x \rightarrow r_p = (0.2)(36.43) = 7.286 \frac{\text{g naph}}{\text{L} \cdot \text{hr}}$$

$$r_{p, \text{total}} = r_p \left(\frac{V_L}{V_r} \right) = 7.286 \left(\frac{50,000}{65,000} \right)$$

$$r_{p, \text{total}} = 5.604 \frac{\text{g naph}}{\text{L} \cdot \text{hr}}$$

e. yield @ SS

$$Y_{p/s, \text{overall}} = \frac{\text{prod form rt } (r_p = q_p C_x)}{\text{total gluc consump rt. } (r_s = \frac{\mu C_x}{Y_{X/S}} + \frac{q_p C_x}{Y_{P/S}})}$$

$$Y_{p/s, \text{overall}} = \frac{q_p}{\frac{\mu}{Y_{X/S}} + \frac{q_p}{Y_{P/S}}} = \frac{0.2}{\frac{0.15}{0.4} + \frac{0.2}{0.25}} = \frac{0.2}{0.375 + 0.8} = \frac{0.2}{1.175} = \boxed{0.170 \frac{\text{g naph}}{\text{g gluc}}}$$

4. Chemostat w/ recycle

Given $\mu = 0.029 \text{ hr}^{-1}$ $C_{X, \text{no recycle}} = 3 \times 10^6 \frac{\text{cells}}{\text{mL}}$

$$q_p = 5 \text{ pg/cell} \cdot \text{day} \rightarrow 5 \times 10^{-12} \frac{\text{g}}{(\text{cell} \cdot \text{day})}$$

$$c = 15$$

a) $\mu = D(1 + a - ac)$ $RIF = \frac{1}{1 + a - ac}$

for recycle expression to remain meaningful

$$1 + a - ac > 0$$

$$1 + a(1 - c) > 0$$

$$1 - a(c - 1) > 0$$

$$a < \frac{1}{c - 1}$$

so $a < \frac{1}{15 - 1} = 0.0714$

$$a_{\text{max}} = 0.0714$$

b) Cell Conc in bio if $a = 0.05$

$$1 + 0.05 - (0.05)15 \rightarrow 1.05 - 0.75 = 0.30$$

$$\text{so } RIF = \frac{1}{0.3} = 3.333$$

$$C_{X, \text{recycle}} = (3.333)(3 \times 10^6) = 9,999,000$$

$$C_x = 9.999 \times 10^6 \frac{\text{cells}}{\text{mL}}$$

c, $D = \frac{\mu}{1 + a - ac} \rightarrow \frac{0.029}{0.30} = 0.0967 \text{ hr}^{-1}$

d) Potent. vol vt of product form
from b = 9.999×10^6 cells/mL

$$9.999 \times 10^6 \frac{\text{cells}}{\text{mL}} \left(\frac{1000 \text{ mL}}{1 \text{ L}} \right) = 9.999 \times 10^9 \frac{\text{cells}}{\text{L}}$$

$$r_p = (5 \times 10^{-12}) (9.999 \times 10^9) = 49.995 \times 10^{-3}$$

round sig fig $5.0 \times 10^{-2} \frac{\text{g}}{(\text{L} \cdot \text{day})} \frac{1000 \text{ mg}}{1 \text{ g}}$

$r_p = 50 \text{ mg Factor } 8/(\text{L} \cdot \text{day})$

5. Cell Conc in Chemostat w/ Recycle

Given $V_0 = 1000 \text{ mL/hr}$ $V = 1000 \text{ mL}$
 $D = \frac{V_0}{V} \rightarrow 1.0 \text{ hr}^{-1}$ $Y_{x/s} = 0.5 \frac{\text{g cells}}{\text{g sub } S}$
 $S_0 = 10 \frac{\text{g gluc}}{\text{L}}$ $\mu_{\max} = 2 \text{ hr}^{-1}$
 $K_s = 1 \frac{\text{g gluc}}{\text{L}}$ $c = 1.5$ $q = 0.7$

a) recycle denom $\rightarrow 1 + \alpha - \alpha C = 1 + 0.7 - (0.7)(1.5)$
 $1.7 - 1.05 = 0.65$

Growth rate $\mu = (1.0)(0.65) = 0.65 \text{ hr}^{-1}$

Residual Sub concn

$$S = \frac{K_s \mu}{\mu_{\max} - \mu} = \frac{(1)(0.65)}{2 - 0.65} = 0.481 \text{ g/L}$$

Cell conc in bioreact

$$C_{x_1} = \frac{Y_{x/s} (S_0 - S)}{1 + \alpha + \alpha C} = \frac{(0.5)(10 - 0.481)}{0.65}$$

$$C_{x_1} = 7.32 \text{ g/L}$$

$$C_{x, \text{recycle}} = c C_{x_1} \rightarrow (1.5)(7.32) = 10.98 \text{ g/L}$$

b. Cell Conc in centrifuge effluent

~~Equation~~ $C_{x, \text{out}} = (1 + \alpha - \alpha C) C_{x_1}$

$$C_{x, \text{out}} = (0.65)(7.32) = 4.758 \text{ g/L}$$